1. Water Jug Problem

Solution:

The Water Jug problem can be solved using a state-space search approach. Starting with both jugs empty, a sequence of fill, pour, and empty operations is performed until the desired state is reached.

Steps:

1. Fill the 3-gallon jug. → (0,3)
2. Pour the 3-gallon jug into the 4-gallon jug. → (3,0)
3. Fill the 3-gallon jug again. → (3,3)
4. Pour water into the 4-gallon jug until it is full. → (4,2)
5. Empty the 4-gallon jug. → (0,2)
6. Transfer the remaining 2 gallons from the 3-gallon jug to the 4-gallon jug. → (2,0)

Thus, exactly **2 gallons of water** are obtained in the 4-gallon jug.

2. Mars Rover Problem

Solution:

The Mars Rover is modeled as a **Utility-Based Agent** that makes intelligent decisions based on environmental conditions and mission objectives.

- **Percepts:** Camera images, terrain information, temperature, obstacle detection.
- **Environment:** Partially observable, dynamic, sequential, and uncertain.
- **Actions:** Move, collect samples, analyze rocks, transmit data, avoid obstacles.
- **Performance Measures:** Mission completion, energy efficiency, data accuracy, and safety.
- **Agent Architecture:** Utility-Based Agent, as it selects actions that maximize mission success.

3. 8-Queens Problem

Solution:

The 8-Queens problem is solved using the **Backtracking Algorithm**. Queens are placed one by one on the chessboard, ensuring that no two queens are in the same row, column, or diagonal. If a conflict occurs, the algorithm backtracks and tries another position until a valid arrangement is found.

A valid arrangement is:

- Queen 1 → Row 1, Column 1
- Queen 2 → Row 5, Column 2
- Queen 3 → Row 8, Column 3
- Queen 4 → Row 6, Column 4
- Queen 5 → Row 3, Column 5
- Queen 6 → Row 7, Column 6
- Queen 7 → Row 2, Column 7
- Queen 8 → Row 4, Column 8

This arrangement ensures that no queen attacks another.

4. OLA Cab Booking Problem

Solution:

The OLA Cab Booking system is best represented using a **Goal-Based Agent**. The agent evaluates the user's preferences, destination, and available cab options to select the most suitable service.

Working:

1. Accept user input (pickup and destination).
2. Display available cab types.
3. Compare user preferences with available options.
4. Select the most appropriate cab.
5. Confirm the booking and provide trip details.

Thus, the Goal-Based Agent helps users achieve their objective of reaching their destination efficiently.

5. Uniform Cost Search (UCS) Problem

Solution:

Uniform Cost Search (UCS) is used to determine the least-cost path from the source node **S** to the goal node **G**. The algorithm expands the node with the lowest cumulative cost at each step.

Example Path:

- S → A (Cost = 1)
- A → C (Cost = 2)
- C → G (Cost = 1)

Total Cost = 1 + 2 + 1 = 4

Therefore, the optimal path found by UCS is:

S → A → C → G

with a minimum cost of **4** units.